Counting Defects: Semilinear Reachability and Resource Bounds for Langton's Ant

Hao Ke

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Abstract

A bounded number of local deviations from the standard Langton highway admits an exact arithmetic description, even when all initial black-cell coordinates are unrestricted. We construct effective semilinear sets of complete finite inputs reaching any supplied defect budget, together with exact event times, poses and endpoint black fields. The construction combines a finite cover by bounded-power words with same-cell successor relations that preserve the entire read chronology. For each fixed finite schema collection, initial-black cost and defect count have piecewise rational-affine spectra. These spectra yield an effective dichotomy between a fixed-cost unbounded-defect ray and a uniform affine supply budget. An exact three-parameter fold–consumer classification gives the sharp inequality $10D \le 9C - 309$. A separate affine-chamber argument supplies polynomial-size positive witnesses and NP membership when the mass and defect budgets are unary. For a restricted segment-source syntax we also construct a polynomial-size quantifier-free physical-realization circuit and a shared factor counter. This backend supports binary linear side conditions and an NP-oracle bound for minimum initial cost. Exact supplied-schema implementations and finite-cover reconstruction accompany the proofs.

1 Problem and main result

A state is (B, p, h), where $B \subset \mathbb{Z}^2$ is the *complete* finite set of black cells, p is the head position, and $h \in \{N, E, S, W\}$. A white read turns right, a black read turns left, the current cell flips, and the ant then moves one unit. Time t denotes the state immediately before update t.

Let W be a fixed cyclic representative of the standard period-104 turn word, and let $\mathcal{L}_{10}$ be its cyclic ten-letter factors. A *contained defect* of a finite forward word is a length-ten factor outside $\mathcal{L}_{10}$. Thus its ending time is at least nine. The contained count and the intrinsic J_{10} count obtained from inverse history can differ in the first nine tests. Their finiteness is equivalent. The retained ten-factor attraction theorem states that a finite-support orbit is eventually standard if and only if it has finitely many such defects [5].

Normalize the head to the origin facing north. For fixed $m \ge 0$ and $N \ge 1$, let $E_{m,N}$ contain the ordered, pairwise-distinct tuples $z = (x_1, y_1, \ldots, x_m, y_m)$ for which the actual ant with exactly these black cells has at least N contained defects. Coordinates have no range restriction.

Theorem 1.1 (All-coordinate fixed-budget synthesis)**.** *There is an effective construction, uniform in the supplied integers m, N, of a Presburger formula defining $E_{m,N}$. The exact time and pose immediately after the Nth defect have an effectively Presburger-definable graph. The complete black field at that endpoint is an effective finite-fibre Presburger family over the initial coordinates.*

In particular, existence, nonexistence, and intersection with a prescribed Presburger coordinate domain are decidable. A finite union handles "at most m black cells." This is not one fixed formula in which arbitrary m, N are additional free variables. The construction may be prohibitively large.

The central difficulty is physical execution: the ant carries an unbounded mutable spatial store, so a short word description must retain every earlier read and every initially black cell. Consecutive same-cell relations solve this problem exactly. A finite ten-factor cover then yields the all-coordinate construction, and unary time-fibre counting yields its resource refinement. Sections 6 and 7 give the supply dichotomy and small-witness complexity theorem.

Symbolic feasibility for ant words has an established history [1]. Presburger descriptions of flat runs are likewise standard in verification [3, 7]. We use these tools to prove complete-field statements for the ant. The ten-factor rigidity and bounded-fold cover are supplied by the companion factor paper [5]; the arithmetic synthesis and resource spectra are the contributions developed here.

2 Consecutive reads without a parity-counting oracle

Fix a parameter tuple θ in a Presburger domain D. Suppose a prescribed kinematic trace is represented by a Presburger function V_θ from a contiguous interval of update times to physical cells. Its domain is either $\{0, \dots, T(\theta) - 1\}$ or $\mathbb{N}$. Let R_θ, L_θ partition its times according to the requested letters. The movements and headings are assumed to satisfy the literal turn rule; Section 3 constructs such traces.

Define

$$C_\theta = \{(s, t) : s < t,\ V_\theta(s) = V_\theta(t)\}.$$

Relational composition uses the same parameter tuple throughout.

Lemma 2.1 (Successor subtraction)**.** *The relation*

$$\mathrm{Next}_\theta = C_\theta \setminus (C_\theta \circ C_\theta) \tag{1}$$

consists exactly of consecutive visits to the same physical cell.

Proof. A pair in C_θ fails to be consecutive precisely when a read time u strictly between its endpoints visits their common cell. This is exactly membership in $C_\theta \circ C_\theta$. No transitive closure is computed. □

Put $\mathcal{T}_\theta = \operatorname{dom} V_\theta$ and

$$\mathrm{First}_\theta = \mathcal{T}_\theta \setminus \operatorname{ran} C_\theta.$$

For a complete initial black predicate $B(\theta, x, y)$, write $\mathrm{Hits}_\theta = \{t : B(\theta, V_\theta(t))\}$.

Theorem 2.2 (Exact realization)**.** *The parameter assignments realizing the prescribed trace are exactly those satisfying*

$$\mathrm{Next}_\theta \cap ((R_\theta \times R_\theta) \cup (L_\theta \times L_\theta)) = \varnothing, \tag{2}$$

$$\mathrm{First}_\theta \cap L_\theta = \mathrm{First}_\theta \cap \mathrm{Hits}_\theta\,. \tag{3}$$

Their set is effectively Presburger-definable.

Proof. Both conditions are necessary by flipping and initial colour. For sufficiency, induct in ordinary time. A first visit has the right initial colour by (3). Otherwise a nearest preceding visit exists: a given time has only finitely many predecessors, even for an infinite trace. Its letter is opposite by (2), so the current physical colour is correct. The requested movement is therefore the actual movement. All integer times are covered. Projection of violations and complementation in D give the Presburger parameter set. □

This quantified form of the established same-cell alternation criterion [1] handles all parameter assignments together. Its advantage is that it avoids assuming an arbitrary definable cardinality has a definable parity.

For a finite feasible trace without a supplied field, the smallest supporting initial black set is

$$B_{\min}(\theta) = V_\theta(\mathrm{First}_\theta \cap L_\theta). \tag{4}$$

For an infinite trace this set may be infinite. Existence on an unspecified colouring must not be mistaken for existence on a finite support.

Theorem 2.3 (Complete endpoint)**.** *For a finite trace, define* $\mathrm{Last}_\theta = \mathcal{T}_\theta \setminus \operatorname{dom} C_\theta$*. On its realization domain the final black field is*

$$B_{\mathrm{out}}(\theta) = \big(B(\theta) \setminus V_\theta(\mathcal{T}_\theta)\big) \cup V_\theta(\mathrm{Last}_\theta \cap R_\theta). \tag{5}$$

It is an effective finite-fibre Presburger family when B is one.

Proof. Unvisited initial cells retain their colour. A last right turn leaves black; a last left turn leaves white. These cases are disjoint and exhaustive. Only finitely many updates occur at each parameter assignment. □

Equation (5) retains every unread initial cell.

3 Unbounded powers and effective resource spectra

Consider a finite bounded-power schema

$$v_0 u_1^{k_1} v_1 \cdots u_s^{k_s} v_s, \qquad k_i \geq 0, \tag{6}$$

with fixed words and heading-restoring u_i. Fixed pieces may rotate heading. An occurrence at phase r and copy index j has time and position of the form

$$t = b_r + c_r k + \ell_r j, \qquad q = a_r + A_r k + d_r j, \qquad 0 \leq j < k_i. \tag{7}$$

The coefficients are fixed integers. There are no products of two unbounded variables. Zero powers and affine reparameterizations are allowed. A final infinite power simply replaces its copy bound by $j \geq 0$.

Theorem 2.2 thus extends fixed-word replay to arbitrarily long parameterized operations, and Theorem 2.3 preserves the complete field. A variable rotating power can first be split by its exponent modulo its rotation order; our implementation instead requires this preprocessing explicitly. Finite-fibre initial Presburger families may include finite unions or XORs of arithmetic segments with fixed strides and variable finite lengths.

Defect end-times in a schema are Presburger: intersect shifted letter-time sets for each allowed factor and subtract their union from eligible times. This treats cross-block windows and zero powers exactly.

Lemma 3.1 (Unary finite-fibre counting)**.** *For an effectively Presburger set $F \subseteq \mathbb{Z}^d \times \mathbb{Z}$ with finite fibres, $\theta \mapsto |F_\theta|$ is effectively rational-affine on the cells of a finite Presburger partition.*

Proof. Apply Presburger quantifier elimination [2]. Split parameter residues and the counted variable modulo a common multiple of the constant moduli. Writing $t = r + Qz$, each remaining atom either constrains the parameters alone or supplies a floor/ceiling affine bound on z. Use a disjoint Boolean partition, then partition parameters by the ordering of the finitely many bounds and their constant-denominator residues. Every nonempty counted cell is an integer interval with two finite rational-affine endpoints. Summing disjoint interval lengths proves the claim. Finiteness rules out unbounded intervals on the applicable parameter cell. □

Theorem 3.2 (Initial-black and defect spectrum)**.** *On the feasible domain of* (6)*, the minimum initial-black cost* $c(\theta) = |\operatorname{First}_\theta \cap L_\theta|$ *and the contained-defect count* $d(\theta)$ *have effective semilinear graphs and are piecewise rational-affine.*

Proof. Apply Lemma 3.1 to the two unary sets of times. Different first-read times refer to different cells, so (4) identifies the first count with initial-black cost. □

This does not assert affine counting for arbitrary planar Presburger fibres; for instance an n by n square has n^2 cells. The counted objects here are one-dimensional time sets. The runtime implements exact time sets and fixed integer thresholds via

$$|F_\theta| > b \iff \exists t_0 < \cdots < t_b \quad \bigwedge_i t_i \in F_\theta.$$

The general symbolic counting-partition algorithm above is a constructive proof, not an optimized backend claimed implemented in this release.

For any supplied finite schema collection and fixed mass bound, unboundedness of its defect spectrum is consequently decidable. If unbounded, a semilinear component gives an affine family of finite witnesses. Such a parameter family is not a fixed infinite counterexample.

4 Why fixed defect counts have complete finite schema coverage

The ant-specific bridge is the existing ten-factor macro theorem [5]. We state its certificate content to make the dependency explicit. All feasible length-150 words with legal ten-factors give a graph with 6,501 vertices and 6,605 edges. Its cyclic components have 104, 24 and 366 vertices. The 366-state component maps to a graph with two loops: a translating word A of length 28 and a heading-reversing word B of length 102. The established physical obstruction is

$$BA^nBA^mB \text{ is infeasible for every } n, m \geq 0. \tag{8}$$

It is an all-parameter same-cell conflict, not an observation at small lengths.

Counting at most two completed long loops and a third incomplete loop yields a finite covering lift with 7,595 vertices and 7,763 edges. Its only cyclic components are simple cycles of lengths $24, 28, 28, 28, 104$. The acyclic condensation supplies bounds of five cycle stages, 956 acyclic updates, and 212 for the sum of visited cycle lengths. Our independent reconstruction regenerates every feasible word, checks every quotient edge, rebuilds the lift and recovers these bounds. The exact arithmetic kernel separately verifies (8) over the entire quadrant.

A finite defect-free word uses at most 149 initial letters before this graph cover applies. Every cyclic visit is a power followed by a partial cycle and possible exit. Therefore it has a schema with at most five powers and at most $149 + 956 + 212 = 1317$ nonpowered letters. The number of ordinary updates is not bounded: all exponents remain free.

Cut a prefix just after each of its first N defects. Applying this cover to the intervening pieces gives a conservative finite schema family with

$$s \leq 5(N+1), \qquad \sum_i |v_i| \leq 2048(N+1) + N. \tag{9}$$

All power words are cyclic phases of W, A, T. This overapproximation may be redundant. Exact arithmetic semantics and exact defect count are subsequently imposed, so no nonphysical word is accepted.

Proof of Theorem 1.1. For each schema in (9), introduce its exponent variables, make the initial field exactly the distinct coordinate list z, and conjoin the realization predicate, exactly N contained defects, and a defect at the last letter. Existentially project exponents and take the

finite union. Every genuine Nth-event prefix is covered; every satisfying assignment is a genuine prefix by Theorem 2.2. Retain the affine time and pose to obtain their graph. Formula (5) gives the black field; different schemas representing the same prefix agree by determinism. □

The finite union supplies a total mathematical decision procedure. In a resource-limited implementation, an unfinished enumeration returns `INCONCLUSIVE`; only complete coverage licenses an emptiness answer.

5 Exact two-fold classification and parametric controls

The kernel obtains the following certificate-assisted proposition. The fixed words are recorded in the source package, with $|B| = 102$ and $A = \mathrm{L}B_{75} \cdots B_{101}$.

Proposition 5.1. *The finite word BA^nBA^m is realizable from some finite support exactly when $n \geq 1$ and $0 \leq m < n$. Its minimum initial-black cardinality is*

$$c(n, m) = 5n + 5m + 47.$$

The feasible domain is computed using (2). Its equality with the displayed integer region is checked by exact Presburger set operations. For the cost, the first B supplies 24 distinct first-left reads, the second supplies 23, and every copy in either A run supplies exactly the five local indices $0, 9, 15, 26, 27$. The full time-set identity is checked parametrically, not extrapolated from sample values. Its disjoint blocks give the formula. The complete declarative time lists are in `PROOF.md`.

A separate old three-sweep rail is accepted for every length $n \geq 1$, with all output rails and four cap cells preserved. For an eastward scanner $(\mathrm{RLLR})^n$, an initial lower rail at $y = -1$, $0 \leq x < 2n$, and an extra black marker (a, b), the exact acceptance domain is

$$n \geq 0 \quad \wedge \quad (b \neq 0 \ \vee \ a < 0 \ \vee \ a \geq 2n).$$

Thus an arbitrary distant initial black cell is not silently ignored. A 10^{60}-copy instance checks the complete endpoint without expanding its ordinary trajectory.

Both successor-subtraction and lexicographic next-read minimization reproduce the double-fold and triple-fold results. They share an exact arithmetic backend and are not independent theorem provers. Finite controls instead use independently written vector rotations and explicit full black fields. An infinite-scanner control rejects an empty finite seed; it succeeds only when an infinite initial black ray is explicitly supplied. This guards against a finite-support mismatch.

6 Supply budgets for a finite schema collection

A supplied finite collection fixes its word descriptions while their exponents remain free. Its resource spectrum distinguishes two possible behaviours.

Theorem 6.1 (Supply dichotomy)**.** *For every finite collection of bounded-power schemas, exactly one of the following alternatives holds, and the alternative is effectively decidable:*

1. *Some fixed initial cost c_0 supports arbitrarily many contained defects. One can extract a schema and an affine ray of exponents $k(n) = k_0 + nv$ with $c(k(n)) = c_0$ and $d(k(n)) = d_0 + n\delta$, $\delta > 0$.*
2. *There are computable rational $\alpha \geq 0, \beta$ such that $d(w) \leq \alpha c(w) + \beta$ for every feasible instance in the collection.*

Proof. Project the semilinear resource graph to $(c, d) \in \mathbb{N}^2$ and compute a finite union of linear sets $b + \sum_j \mathbb{N} g_j$. A generator with zero cost coordinate and positive defect coordinate gives the

first alternative. Conversely, if cost stays bounded while defects grow, positive-cost generators have bounded multiplicities in some component, so a positive vertical generator is necessary. When there is none, set

$$\alpha = \max\big(\{0\} \cup \{g_{j,d}/g_{j,c} : g_{j,c} > 0\}\big), \qquad \beta = \max_b(b_d - \alpha b_c).$$

The inequality holds at each base and is preserved by every generator. Use zero constants for the empty spectrum. To extract the first alternative's exponents, fix its c_0 in the full (k, c, d) graph. An unbounded component has a generator with positive d coordinate, giving the asserted affine ray. The minimal seeds are computed by first-left reads for each parameter. □

The ray describes a family of finite inputs. The constants in the other alternative belong to the specified collection. When a collection has unbounded defects and no vertical ray, its least supply for at least N defects is $\Theta(N)$: the affine budget gives the lower bound, and any positive-defect generator gives the upper bound.

6.1 Three independent fold and consumer parameters

Use the inherited 28-letter word A, 102-letter connector B, and

$$P = \mathrm{RLRRRRLRRRRLLLLRLRRL}.$$

From $(-1, 0)$ facing west consider

$$w(a, b, r) = B[9:]A^a B A^b P^r, \qquad a, b, r \geq 0.$$

Here $B[9:]$ deletes nine initial letters. The input field is synthesized from first reads, so the three exponents are independently quantified.

Theorem 6.2 (Exact fold–consumer spectrum)**.** *The word is feasible exactly when*

$$a \geq 1, \qquad 0 \leq b < a, \qquad 0 \leq r \leq b + 2.$$

On this domain, its minimum initial-black cost and contained-defect count are

$$C = 46 + 5a + 5b, \qquad D = \begin{cases} 0 & r = 0, \\ 9r - 3 & r \geq 1. \end{cases}$$

Its resource image is exactly

$$\begin{aligned} &(51, 0) + \mathbb{N}(5, 0), \\ &(51, 6) + \mathbb{N}(5, 0), \\ &(51, 15) + \mathbb{N}(10, 9) + \mathbb{N}(5, 0). \end{aligned}$$

Consequently $10D \leq 9C - 309$*, with sharp slope and intercept. For a complete input of mass* $m \geq 51$*, the largest possible count within this family is*

$$D_{\max}(m) = 15 + 9 \left\lfloor \frac{m - 51}{10} \right\rfloor.$$

Proof. Compile every phase of the five blocks into its exact position–time relation. Same-cell successor alternation gives a Presburger feasible set; the supplement checks its equality with the displayed region, with no bound on a, b, r. First-left times are partitioned into disjoint residue intervals. Each proposed interval is accepted only after exact set equality with the corresponding fibre, so its integer length counts all first-left reads. Their sum is $46 + 5a + 5b$. The shifted

ten-letter matcher includes all junction windows and zero powers, and its time-set count gives the stated D. These are exact symbolic identities in the supplied finite certificate.

For $r = 0, 1$, put $b = 0, a = 1 + s$ to obtain the first two components. For $r \geq 2$, put $r = q + 2$; its least feasible b, a are $q, q + 1$. Increasing a supplies all extra multiples of five, yielding precisely the third component. The inequality holds on all three components and is equality at $a = b + 1, r = b + 2$ for every $b \geq 0$, proving both sharpness claims. For the mass optimum take $b = \lfloor (m - 51)/10 \rfloor$, $a = b + 1, r = b + 2$; place any remaining black cells outside the finite read footprint. □

This extends the symmetric producer family used in the factor paper to all independently sized folds and finite consumer trains. Its terminal black mass is $67 + 9(a + b) + 4r$, so initial supply and final mass measure different resources.

7 Small positive witnesses and unary NP membership

Semilinearity provides a decision procedure. The fixed geometry of the cycle library also gives a quantitative size bound on a positive certificate. Let $E(m, N)$ mean that some complete seed of at most m black cells produces at least N contained defects, with all coordinates unrestricted. Put

$$s = 5(N+1), \quad K = 2570(N+1), \quad d = s+2m, \quad H = 2^{24}(K+d+1), \quad R = 64((d+1)!H^d+1).$$

Theorem 7.1 (Small witnesses)**.** *If $E(m, N)$ holds, it has a witness whose initial coordinates and schema exponents have absolute value at most R. Its Nth defect occurs within $K(R + 1)$ updates. In particular, $\log R = O((m + N + 1) \log(m + N + 2))$. With m, N in unary, $E(m, N)$ belongs to NP.*

The radius is existential: it produces a possibly different input with the same budgets. A specified input may put its next obstacle arbitrarily far away.

7.1 A polynomial verifier for a compressed physical run

The canonical cover has at most s powers and $2048(N + 1) + N$ literal letters. There are at most K phase types. A powered type has position $A_i(k) + jD_i$ and affine time, with $0 \leq j < k_i$. The nonzero strides, after rotations, are

$$(\pm 2, \pm 2), \quad (\pm 2, 0), \quad (0, \pm 2), \quad (\pm 4, 0), \quad (0, \pm 4).$$

For a proposed phase occurrence, its read colour is initial membership XOR the parity of every earlier type visiting the same cell. Each nonzero-stride type visits a given cell at most once. Solve $A + jD = B + \ell E$ for each pair of types. Nonparallel strides give at most one integral pair, with determinant dividing 16. Parallel strides yield a bounded interval in one residue class of j, of modulus at most two. A source point or literal type gives at most one hit.

Different blocks have their fixed chronological order. Within one powered block, intersecting phases have $\ell = j + c$ for a constant c; the sign of $|u|c +$ old phase $-$ current phase decides precedence. Thus all previous flips are included. Split the current index by its hit moduli, sort interval starts and one-past-ends, and check the parity on each nonempty interval against the requested bit. Induction on ordinary update time proves that these checks are necessary and sufficient for execution on the complete supplied field.

There are $O(K(K+m))$ pair and point queries and $O(K(K+m) \log(K+m)+m^2)$ comparisons, including source distinctness. All operate on binary integers of polynomial length. A word summary stores length, first and last nine symbols, and contained-defect count; concatenation checks only crossing windows and binary powering is logarithmic. In this library all powered words are cyclically ten-legal and have length at least 24, so the count is fixed once every exponent is classified as zero or positive.

7.2 Linear chambers through successful certificates

Let θ list the independent exponents and the coordinates of $m' \le m$ distinct initial black cells; its dimension is at most d. At one successful witness fix $\theta \bmod 64 = \rho$ and write $\theta = 64z + \rho$. Follow the verifier and record every comparison with its witnessed outcome, including endpoint ordering, equalities, source distinctness, and exponent zero/positive cases. Strict integer inequalities become non-strict inequalities by a unit shift. For a failed equality record its strict sign.

All modular tests are now constant. Determinant denominators divide 16, point-membership denominators divide four, and the parallel-index and parity splits each divide by at most two; the accumulated divisors divide 32. Floors and ceilings are affine in z, with only the constant term rounded. Every integer solution to the resulting chamber follows the same comparison path, hence preserves physical feasibility and exact defect count.

For completeness, the coefficient envelope can be followed explicitly. Set $H_0 = 2^8(K + d + 1)$. After the substitution, origin variable coefficients are at most 256 and their constants at most $K + 252s$. Origin differences fit $2H_0$. Determinant numerators fit $16H_0$ and their quotients fit $16H_0 + 1$. Parallel-hit endpoints fit $4H_0 + 16$; after parity splitting and successors all endpoints fit $16H_0 + 32$. Minima and maxima select one existing form. Endpoint comparisons and collinearity guards fit $128H_0 + 128$. Within-body origin differences cancel all parameters. Thus every integer coefficient and right-hand side is bounded by $256H_0 \le H$.

Proof of Theorem 7.1. The integer small-solution theorem gives a solution to every nonempty n-variable chamber of coefficient height $a \ge 1$ with coordinates at most $(n+1)!a^n$ in absolute value [6]. Apply it with $n \le d, a \le H$, then substitute $\theta = 64z + \rho$. This proves the stated R. The source still has distinct cells, the powers remain nonnegative, and its word still ends at the Nth defect. Summing at most K phase types gives the duration bound. Taking logarithms yields the bit bound.

For unary budgets guess the schema, source coordinates and exponents inside these bounds. They use $O((m + N + 1)^2 \log(m + N + 2))$ bits. Check the conservative schema-size limits, library membership, exact physical execution and defect count using the preceding algorithm. Every accepted certificate is a genuine run, and the finite cover plus small-solution argument supplies one for every positive instance. This proves NP membership. □

The chamber argument applies to the canonical library with free initial coordinates. A supplied affine input family with larger coefficients requires its own coefficient-height bound.

Corollary 7.2. *The minimum initial mass $\mu(N)$ needed for at least N contained defects can be computed in deterministic time $2^{O(N^2 \log(N+2))}$.*

Proof. The word $(\mathrm{LRRL})^q$ has minimum supply $2q$ and $\max(0, 4q - 9)$ defects, so $\mu(N) \le 2\lceil (N + 9)/4 \rceil = O(N)$. Enumerate the bounded positive certificates for successive masses through this upper bound. The preceding bit-size estimate gives the claimed time. □

Short schemas and small integer solutions are established tools for flat counter systems [7]. Here the mutable-field interval verifier supplies the additional ant-specific execution argument.

8 A polynomial-size quantifier-free decision backend

The general Presburger construction supports arbitrary finite-fibre source predicates and complete target equality. For the modal schemas in the finite-defect cover, a narrower syntax admits a polynomial-size existential encoding. This provides an executable decision backend and a second route to the small-witness conclusion.

Let

$$\mathcal{D} = \{(\pm 2, \pm 2), (\pm 2, 0), (0, \pm 2), (\pm 4, 0), (0, \pm 4)\}.$$

Every powered word must restore heading and have full drift in $\mathcal{D}$. The source is a finite XOR of segments $a + jv$, $0 \le j < L$, with $v \in \mathcal{D}$, and singletons. Anchors, lengths and exponents may be affine expressions in shared integer parameters; lengths and exponents are constrained nonnegative. Explicit m-point inputs additionally require pairwise distinct coordinates. No coordinate or repetition bound is imposed.

Theorem 8.1 (Quantifier-free physical realization)**.** *For this syntax, a polynomial-time construction produces a quantifier-free linear-integer/Boolean circuit whose existential satisfiability is equivalent to literal execution of the complete prescribed word. Exact contained-defect counts, the final defect flag, endpoint time and pose, and any polynomial number of endpoint-colour queries have polynomial-size circuits as well.*

Proof. Represent each phase of a power by the injective read stream $a_i + jd$, $0 \le j < k$, with its fixed required colour. Its actual colour is the XOR of the source and every earlier planned write at that point. For streams in different blocks, temporal order is fixed. Within one power of period P, a phase-r occurrence at index $j+s$ precedes phase i at index j exactly when $Ps+r < i$. This retains the intra-loop chronology rather than grouping all occurrences by phase.

For a source or write stream $b + \ell v$, solve $a + jd = b + \ell v$ with its finite index bounds. Nonparallel strides give at most one hit by the determinant formula. Parallel strides give a guarded arithmetic interval in j with modulus $|v_\alpha|/\gcd(|d_\alpha|, |v_\alpha|)$, which is always 1 or 2 for $\mathcal{D}$. Splitting $j = 2z + e$ leaves a guarded ordinary interval. Floors and exact divisibility use only fixed integer divisors and therefore have constant-size linear defining constraints.

On a target residue interval $[0, K]$, the XOR of these interval contributions is a step function. It has the required constant colour exactly when its value at zero is correct and every active interior boundary has even total jump parity. Check this at each candidate interval start and one-past-end, including all coincident boundaries. For q contributions, the equality and parity tests use $O(q^2)$ shared wire operations; no ordering of symbolic endpoints is enumerated. With R read streams and M source terms the parity stage has size $O(R(R+M)^2)$, apart from shared affine expressions. Induction through the finite planned word proves that these colour tests are sufficient as well as necessary for physical execution.

For lexical counts, retain the preceding at most nine turns, their length, the defect count and the final-event flag as shared registers. For a power u^k, at most $\lceil 9/|u| \rceil$ copies precede suffix stabilization; later copies contribute a fixed cyclic defect count. Constant-size guarded transfers select the short cases and the stabilized case. The $k = 0$ branch preserves the entire incoming state. Sharing these transfers between blocks gives a circuit linear in spelled length and block count, instead of a Cartesian list of zero/positive choices.

Finally, an endpoint-colour query XORs initial membership with membership in every read stream. This retains all unqueried remote cells. Named gates and linear definitions preserve polynomial size throughout and prove the theorem. □

Corollary 8.2 (Unary budgets with binary side conditions)**.** *For explicit m coordinate pairs, unary $N \ge 1$ and a quantifier-free linear-integer side condition Q of binary length b, existence of a complete m-black source whose Nth contained defect satisfies Q is in NP. Positive witnesses have polynomial bit length in $m + N + b$. For unary N, the minimum initial cost $\mu(N)$ is in* $\mathrm{FP}^{\mathrm{NP}}$.

Proof. The inherited complete cover supplies a schema with at most $5(N+1)$ powers and $2048(N+1)+N$ fixed letters. Its powered bodies are phases of W, T, A, with strides in $\mathcal{D}$. Guess this polynomial-length schema, apply the theorem to the m distinct source points, and require exactly N defects with a bad last letter and condition Q. Guessing Boolean branches reduces the circuit to polynomially many linear integer constraints. The integer small-solution bound used in Section 7 supplies polynomial-bit witnesses, which can be checked in polynomial time. This uses the complete physical cover; satisfiability of a graph relaxation alone would not suffice.

The finite row construction $(LRRL)^L$ gives $\mu(N) \le 2\lceil (N+9)/4 \rceil$. A finite prefix can be padded by unread black cells outside its footprint, so the NP query for exactly m points decides $\mu(N) \le m$ without additional side conditions. Binary search over this $O(N)$ range needs $O(\log(N+1))$ NP-oracle queries of polynomial unary-N size. No value of μ or divergence statement is computed here. □

The new backend rejects arbitrary quantified source predicates, zero-displacement powers and complete-field target equality. Those tasks remain with the more general compiler. Polynomial circuit size does not imply a polynomial deterministic solver, and the NP statement is not a claim for binary-encoded N.

The bundled checks include 312 all-integer intersection equivalences, 544 parameter instances across 34 schemas, 278 complete valid endpoints, 175,175 endpoint-bit comparisons, exact huge-coordinate identities and eight invalid-input controls. Z3 and ISL/GMP are explicit trust dependencies; native UNSAT responses are not independently checked proof objects. Historical timing measurements show a roughly 128-fold gain for the factor-counter construction at 16 independent powers, but only about 1.8–1.9 on three supplied-schema decision examples and a slowdown on one small example. These are distinct tasks; no global synthesis speedup is claimed and timing is not a deterministic replay target.

9 Fixed budgets and eventual convergence

Define the root-cost sequence

$$\mu(N) = \min\{c(w) : w \text{ is finite feasible and } d_{10}(w) \ge N\}.$$

A row scanner $(\mathrm{LRRL})^q$ uses $2q$ initial black cells and has $\max(0, 4q-9)$ defects. Hence

$$0 \le \mu(N) \le 2\left\lceil \frac{N+9}{4} \right\rceil.$$

Theorem 1.1 makes every fixed-mass query decidable, so this finite upper bound proves that μ is a total computable nondecreasing function. Corollary 7.2 supplies a quantitative computation bound for this function.

Corollary 9.1. *If $\mu(N) \to \infty$, then the finite-support Highway Conjecture holds.*

Proof. An m-black input has first-black cost at most m on every prefix. Divergence of μ bounds its defect count. The ten-factor attraction theorem then makes its future standard. □

The premise is unproved and stronger than the exact HC formulation. Indeed, HC asks for $\bigcap_N E_{m,N} = \varnothing$ for every m, whereas a coordinate-independent event budget asks that some individual $E_{m,N}$ be empty. Nested semilinear sets $\{a : a \ge N\}$ show that these quantifiers cannot be interchanged. A family of increasingly remote finite witnesses need not contain one infinite counterexample.

The synthesis theorem fixes the requested event count and quantifies over all coordinates. The supply dichotomy instead fixes a finite schema collection and allows unbounded repetitions. Keeping these two inputs explicit gives a way to study which physical mechanisms can produce many defects at low cost.

10 Reproducibility

Run `python3 defect_synthesis/reproduce.py`. The self-contained supplement contains two exact relation compilers, independent finite-state controls, the finite-cover reconstruction, the

complete three-parameter resource identity and fixed-schema query optimizations. Python 3.10+, a C++17 compiler and the system ISL/GMP and Z3 shared libraries are required. The arithmetic backend performs exact integer set operations [4].

The replay checks supplied-schema identities and the finite cover. The universal nonzero-budget disjunction is specified constructively; it is too large to be part of the executed replay. The small-witness theorem is supported by the interval and chamber proof in Section 7. The source manifest and review record distinguish these analytic arguments from computational checks.

For repeated integer threshold queries, the optimized kernel represents a unary time set by its strict-order relation. Its mth relation power witnesses $m+1$ distinct times, so binary powering reduces a fresh threshold query to $\lfloor \log_2 m \rfloor + \mathrm{popcount}(m) - 1$ compositions for $m \geq 1$. A shared decision graph also evaluates the 84 allowed ten-factors with 183 mask intersections instead of 840. Exact set-equality controls compare the optimized queries with their baseline semantics. These improvements preserve the mathematical schema cover and the complete endpoint field.

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